

ADHESIVE DENTISTRY REVIEW

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MUSC -JBECDM

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The Best Preparation

NO PREPARATION!!!

Caries Management - Patient
Education & Remineralization!

Surgerize ONLY when we can't
remineralize!

Before you Prepare a tooth....

Check for tooth Vitality!!!!



Keys to **CRITICAL THINKING**

Stop and Think

R

ecognize Assumptions

E

valuate Information

D

raw Conclusions

Plan of Action

Isolation



Mission Critical

STEP 1 >

Remove Biofilm

STEP 2 >

Etch

STEP 3 >

Disinfectant

STEP 4 >

Prime
and
Bond

4

Steps

BioFilm

Plaque and other organic material that resides on the tooth. This will inhibit an effective bond to any type of tooth structure.



Smear Layer

Denatured cutting debris
formed by instrumentation of
dentin.

It is **welded** to the tooth
surface. (6 MPa)

STEP #1

Cleaning the enamel removes the biofilm and smear layer on these surfaces, enhancing the overall bonding process.



STEP 2 Enamel

TOTAL ETCH, 37% phosphoric acid
15 SECONDS – WASH, GENTLY DRY

On enamel, 37 % phosphoric acid
etching helps remove the smear
layer and create a
microscopically rough surface.



STEP 2

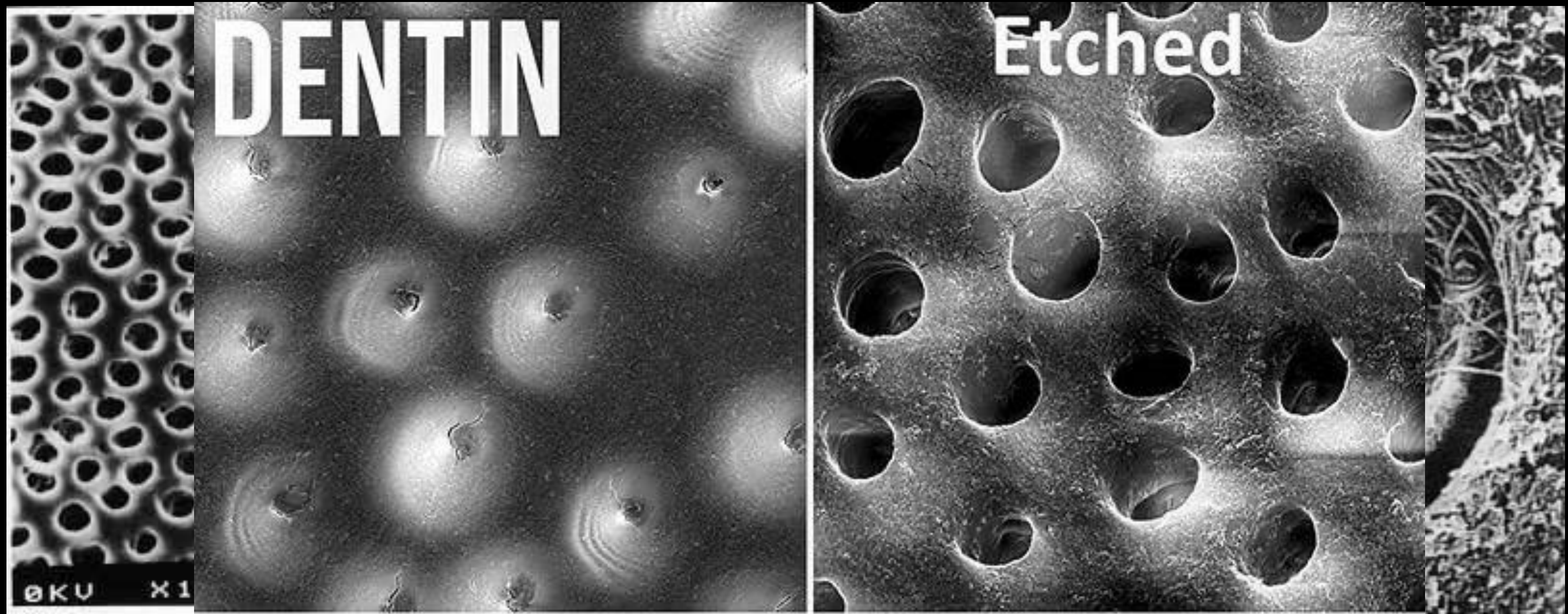
Dentin

TOTAL ETCH, 37% phosphoric acid
15 SECONDS – WASH, GENTLY DRY

Etching removes the smear layer (good) but excessive etching removes excess calcium from the hydroxyapatite (bad). You must etch properly.



Exposed



Instagram



pH~0.7

STEP 3

Apply FiteBac to enamel and dentin.
A methacrylate QAS will penetrate the dentin, remove the smear layer and mineralize the dentin to protect the dentin and kill bacteria on contact. This will enhance the long-term bond.





One of these two techniques **MUST** be employed!!!

WITH DIRECT COMPOSITE RESTORATIONS UTILIZING TOTAL ETCH ADHESIVES / TECHNIQUES

1. Isolate tooth and prepare the cavity. Rinse with water and air dry.
2. Etch enamel and dentin for 15 seconds with phosphoric acid according to manufacturer's instructions. Rinse with water and dry, ensuring to leave surface slightly moist (non-desiccated).
3. Shake bottle 2 - 4 seconds just before use. Apply **FiteBac® Antimicrobial Cavity Cleanser** generously to the prepared surface using brush or absorbent pellet.
4. Gently air dry to remove puddled solution, or dry using a new absorbent pellet, leaving site moist.
5. Apply your bonding agent of choice according to manufacturer's instructions.

WITH DIRECT COMPOSITE RESTORATIONS UTILIZING SELF-ETCH TECHNIQUE

1. Isolate tooth and prepare the cavity. Rinse with water and air dry, ensuring to leave surface slightly moist (non-desiccated).
2. Shake bottle 2 - 4 seconds just before use. Apply **FiteBac® Antimicrobial Cavity Cleanser** generously to the prepared surface using brush or absorbent pellet.
3. Gently air dry to remove puddled solution, or dry using a new absorbent pellet, leaving site moist.
4. Apply your bonding agent of choice according to manufacturer's instructions.

STEP 4

Bonding to Enamel and Dentin

Apply two separate coats of All Bond Universal and scrub with a microbrush for 10 -15 seconds per each coat.

Air dry after second coat for 10 second and light cure for 10 seconds.



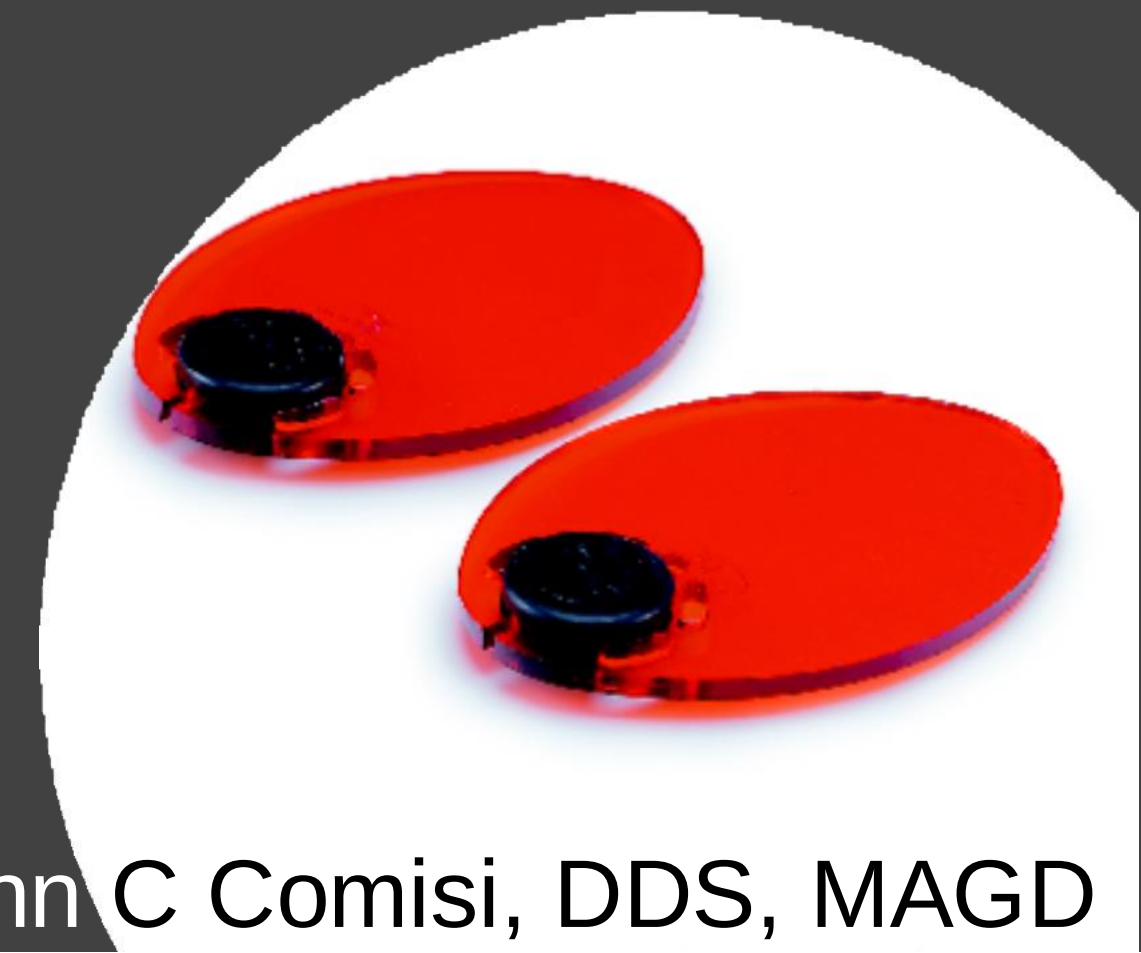
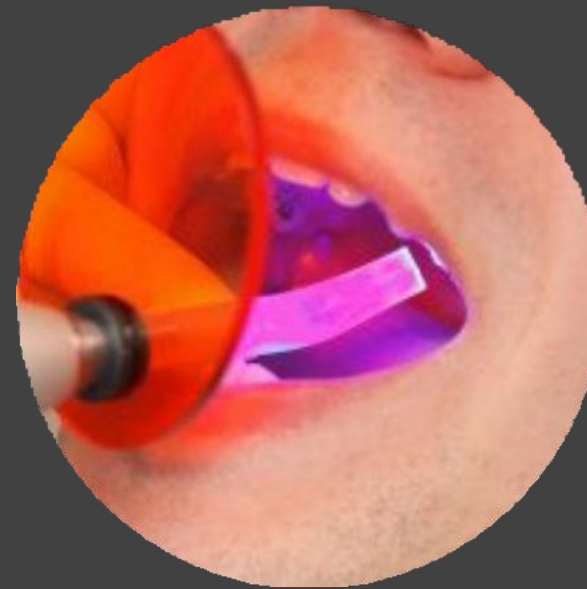
LIGHT CURING!!

If energy is not available, the components will NOT cure!!

Pre-Exposure Checklist for Light Curing Unit

Safety precautions

- Eye safety is required

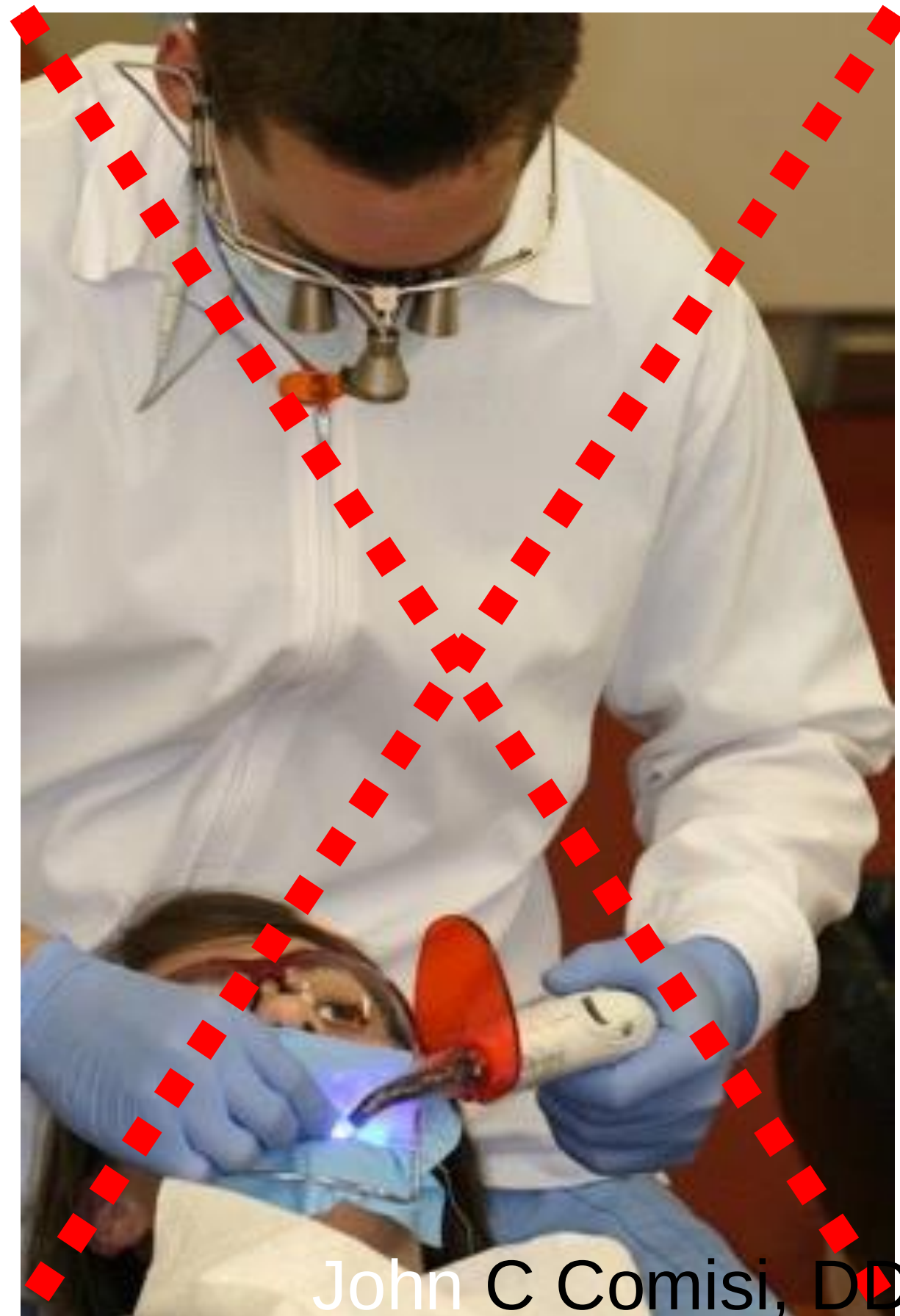


This is a Recipe for Failure

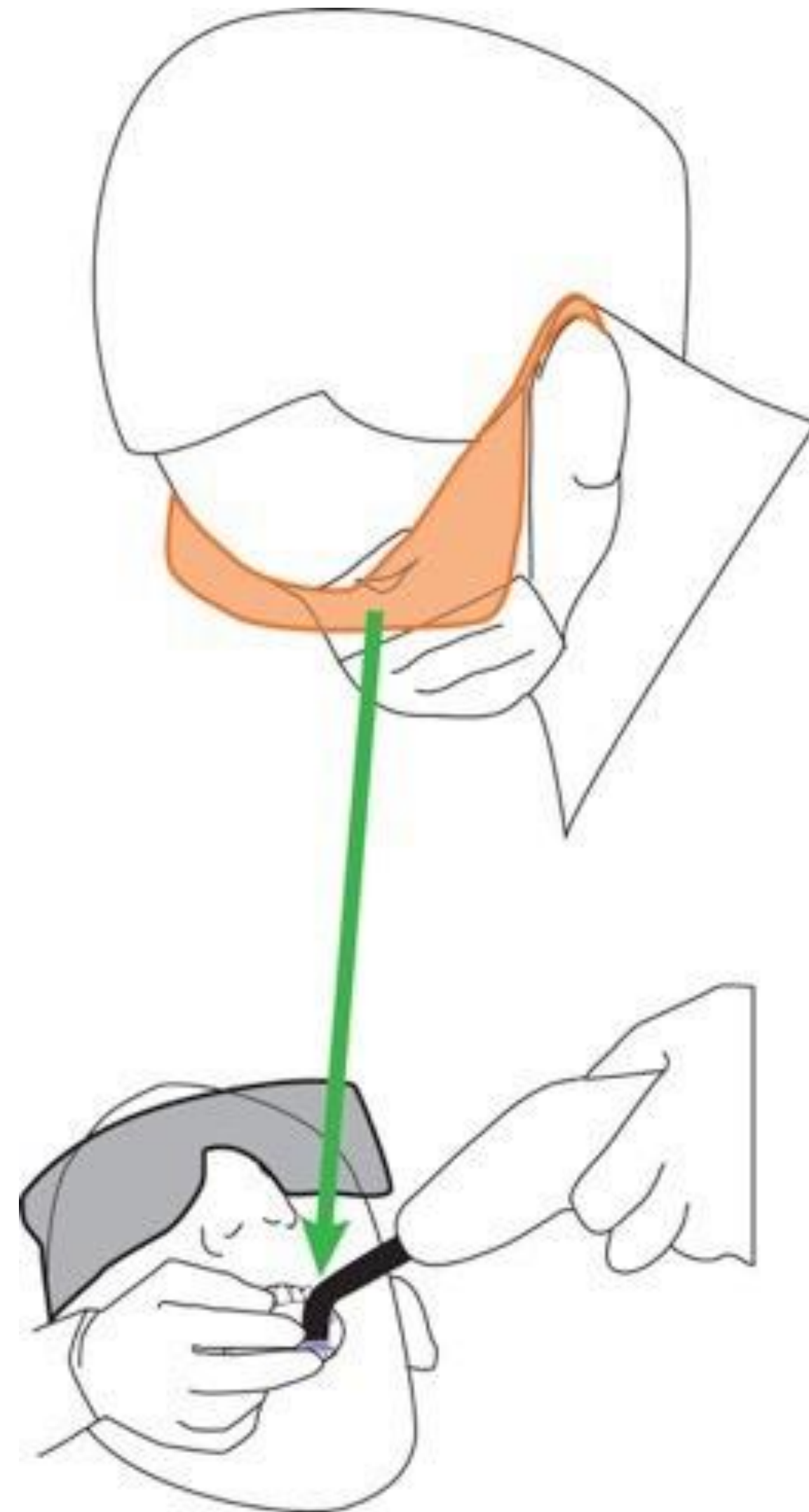


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Must Use Eye Protection



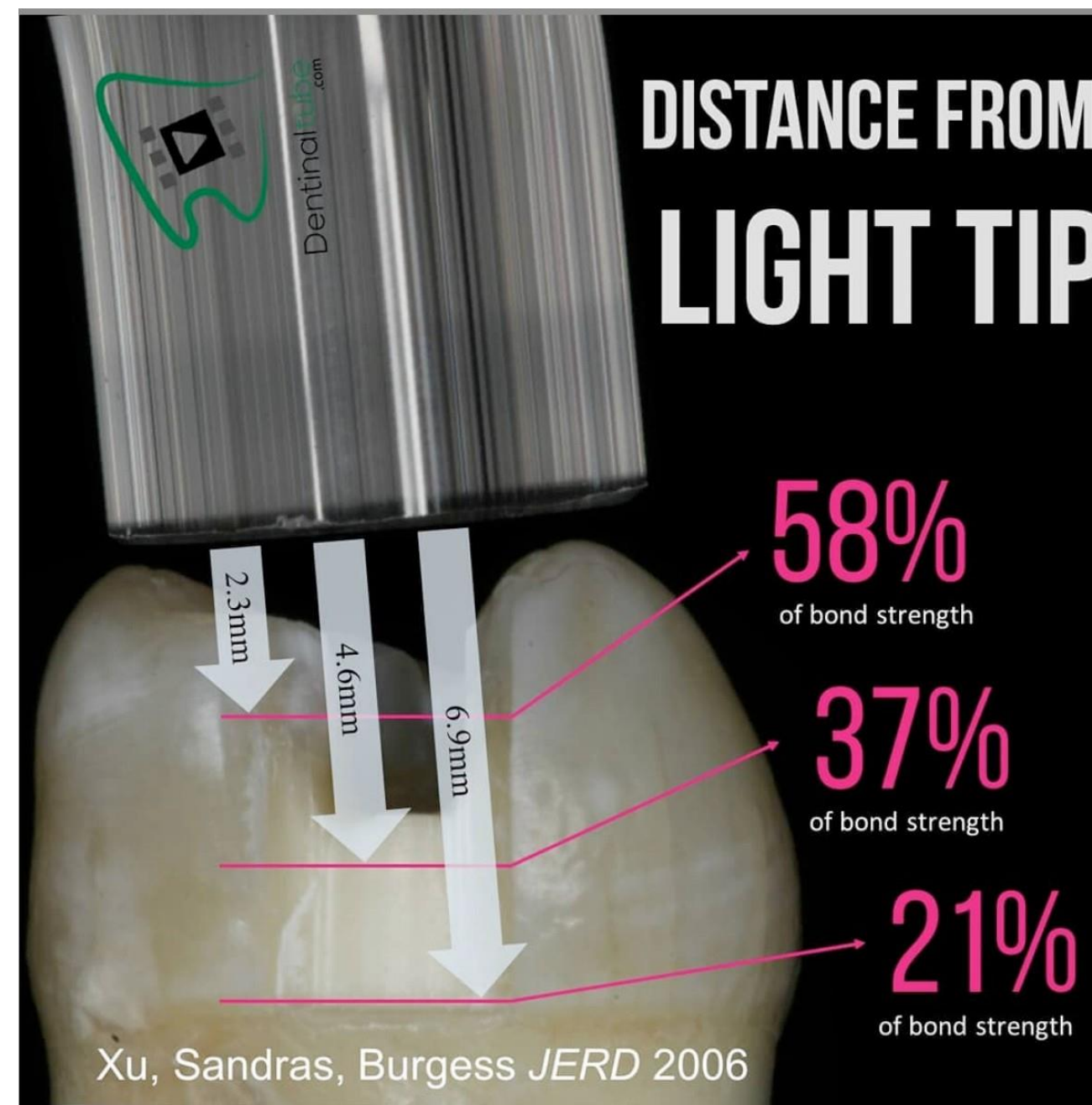
- **Wear Eye Protection**
- **Stabilize tip**
- **Watch** when light curing



Optimizing Clinical Effectiveness

Curing light needs to be close to and effectively cover the entire restoration, or need to overlap the curing of the restoration to be sure to achieve maximum possible cure. The deeper the prep, the more light energy is needed.

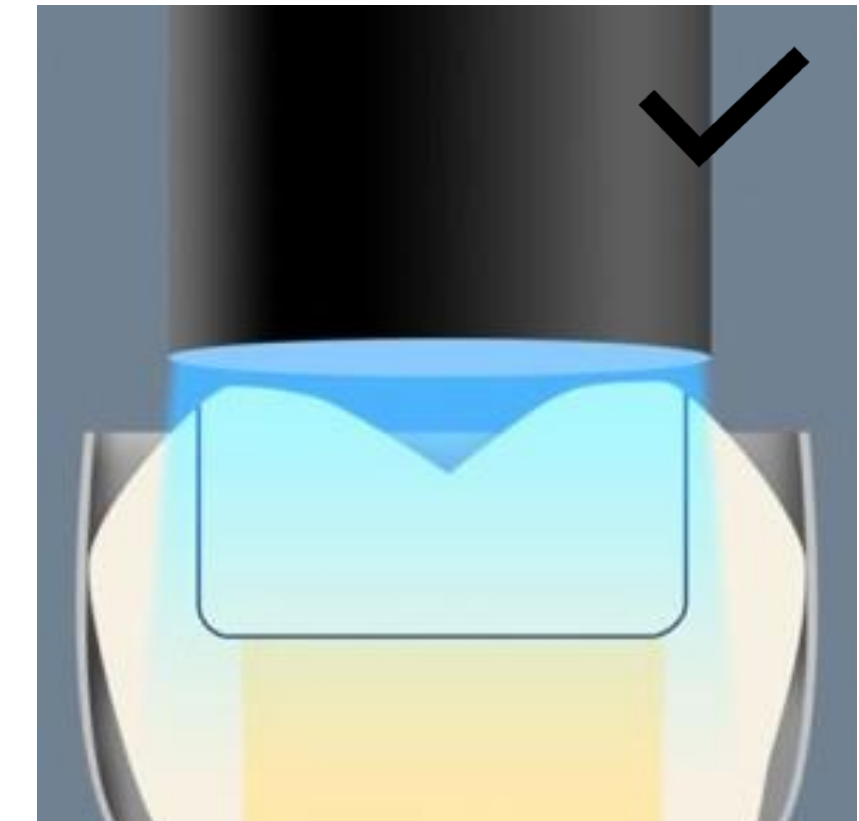
A Plastic Sleeve on the curing light is optimally needed, make sure seam of plastic sleeve does not cross over the curing light aperture.



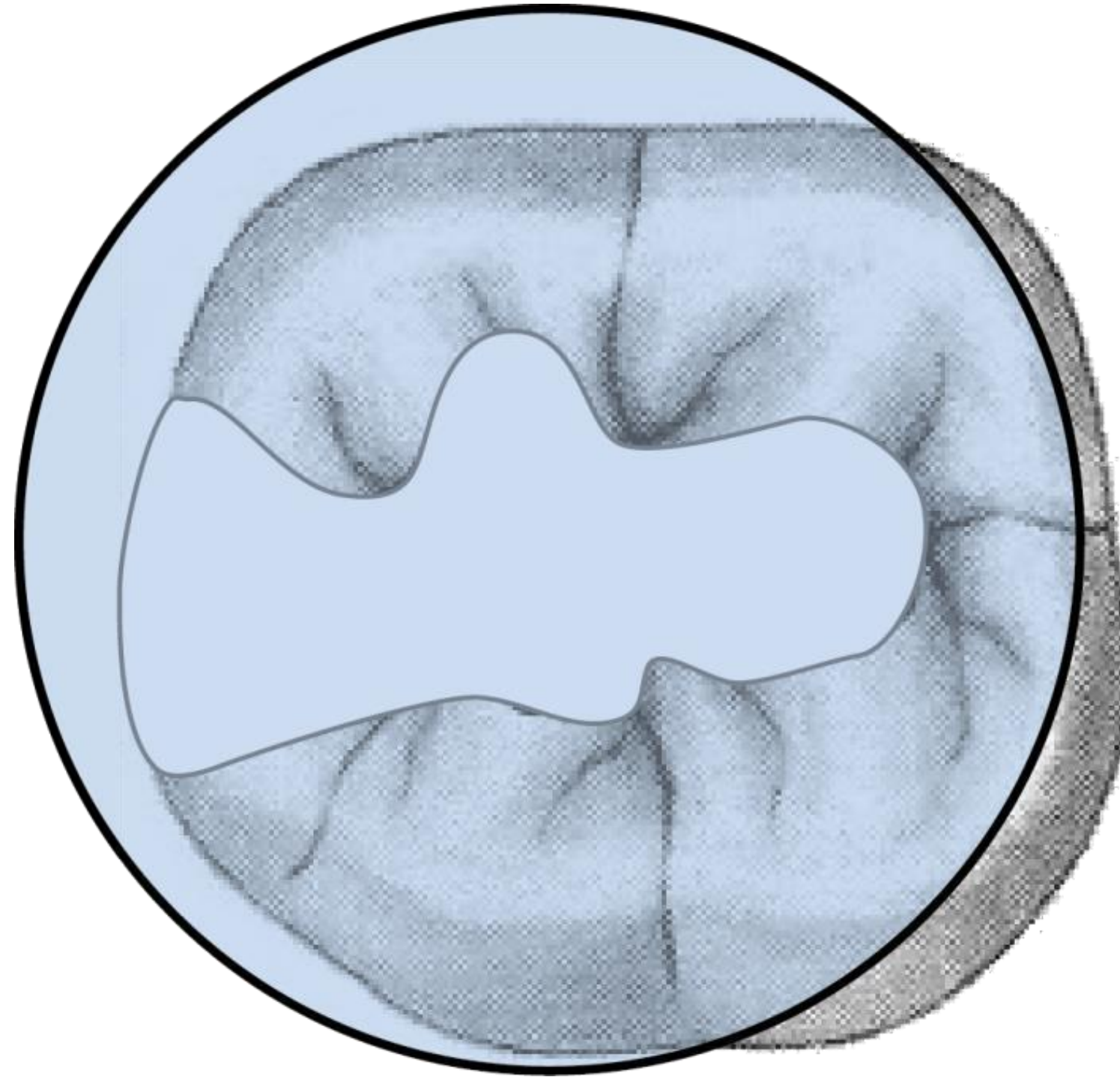
Tip Position

Optimizing Clinical Effectiveness

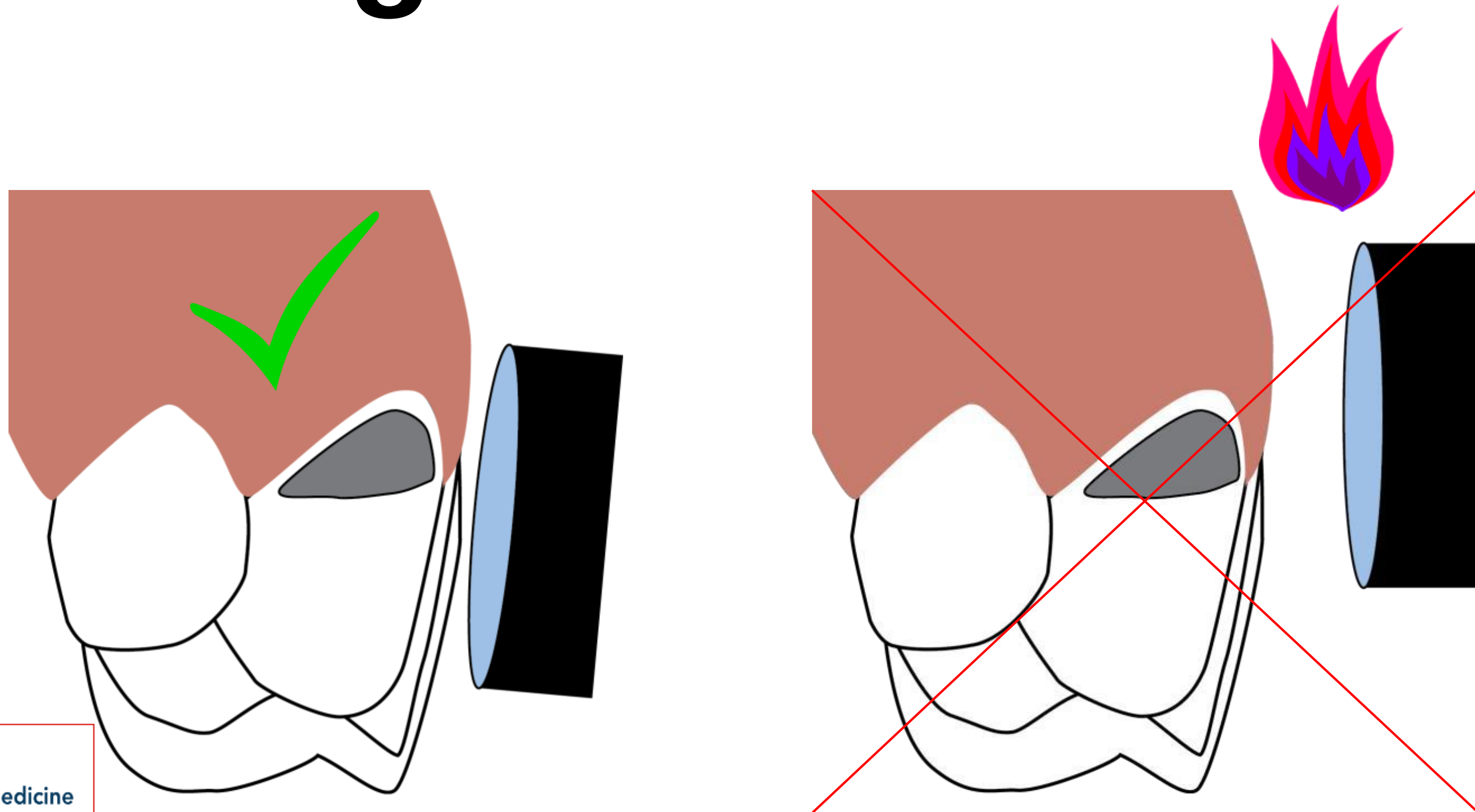
- Factors during exposure
 - Tip positioning (create no shadows)
Minimize tip-to-target distance.
 - Direct observation of target during exposure. (LOOK at the restoration with eye protection!!!)
 - Keep the light STABLE on the tooth.
 - Adjust exposure duration for decrease in power of light.
 - If curing longer, air cool the tooth with air/water syringe.



Cover Restoration with Tip



Position Light To Minimize Thermal Damage To Soft Tissue



Review:

- What are the **4** steps involved in the bonding process and what is the purpose of each?
Biofilm Removal, Etch, disinfect, prime/bond
- Review the smear layer and its importance in bonding.
Debris that is fused to the dentin. Must be removed or modified to bond well.
- Know the mechanism by which composites bond to teeth.
Micro-mechanical/mineral exchange (with use of functional monomers).
- Which bonding system do we use here at MUSC and what are the steps for application?
All Bond Universal.
Steps: Preppies, Etch (15 secs, wash and gently dry), Disinfect with FiteBac, apply two coats (scrub 10-15 seconds each coat) which enables prime/bond to occur, air dry 10 seconds after second coat then light cure 10 seconds).